

06/07/20

S T Q Q S S D

Atividade 1-113

① Área da região delimitada pelas curvas:

$$y = e^{-x}, y = x+1, x = -1$$

$$\text{Intervalo} = [-1, 0]$$

$$\text{Área} = \int_{-1}^0 f(x) dx - \int_{-1}^0 g(x) dx \rightarrow \int f(x) dx = \int e^{-x} dx = [-e^{-x} + C]$$

$$\int g(x) dx = \int (x+1) dx = \left[\frac{x^2}{2} + x + C \right]$$

ALTERNATIVA E

$$\text{Resposta} = \frac{1}{2} - \frac{1}{e} \text{ u.a.}$$

$$\begin{aligned} -e^{-x} &\rightarrow -e^0 = -1 \rightarrow -1 - \frac{1}{e} \\ \frac{x^2}{2} + x &\rightarrow \frac{0^2}{2} + 0 = 0 + \frac{1}{2} = \frac{1}{2} \end{aligned}$$

② Encontrar a área na figura.

$$\int_1^2 \frac{1}{x} dx = \ln(2) - \ln(1) = \ln(2)$$

ALTERNATIVA C

$$\ln(2) - \ln(1) + \frac{1}{2} - \frac{1}{2} = \ln(2) \text{ u.a.}$$

③ Calcular o comprimento da curva dada por $y = \frac{x^4}{4} + \frac{1}{8x^2}$, $1 \leq x \leq 2$

$$S = \int_1^2 \sqrt{1 + [f'(x)]^2} dx \rightarrow f'(x) = \frac{4x^3}{4} - \frac{x^{-3}}{4} = \left[x^3 - \frac{1}{4x^3} \right]$$

$$\left[x^3 - \frac{1}{4x^3} \right] \left[x^3 - \frac{1}{4x^3} \right] = x^6 - \frac{x^2}{4x^3} - \frac{x^2}{4x^3} + \frac{1}{16x^6} = x^6 - \frac{1}{2} + \frac{1}{16x^6} \rightarrow +1 = x^6 + \frac{1}{16x^6} + \frac{1}{2}$$

$$S = \int_1^2 \sqrt{x^6 + \frac{1}{16x^6} + \frac{1}{2}} dx = \frac{1}{4} \int_1^2 \sqrt{16x^6 + 1 + 8} dx \longrightarrow$$

$$\int \frac{4x^6 + 1}{x^3} dx \rightarrow 4 \int x^3 dx + \int \frac{1}{x^3} dx \rightarrow \frac{4}{4} x^4 - \frac{1}{2x^2} \longrightarrow$$

$$\frac{x^4}{4} - \frac{1}{8x^2} + C \rightarrow 2 \rightarrow 4 - \frac{1}{32} = \frac{32}{8} - \frac{1}{32} = \frac{128}{32} - \frac{1}{32} = \frac{127}{32}$$

$$\frac{x^4}{4} - \frac{1}{8x^2} \rightarrow 1 \rightarrow \frac{1}{4} - \frac{1}{8} = \frac{2}{8} - \frac{1}{8} = \frac{1}{8}$$

$$\frac{127}{32} - \frac{1}{8} = \frac{127}{32} - \frac{4}{32} = \frac{123}{32} \text{ mC} \quad \boxed{\text{ALTERNATIVA D}}$$

4) Calcular o comprimento:

$$\begin{cases} x = t^3, & 1 \leq t \leq 3 \\ y = t^2, & \end{cases} \quad \begin{cases} x'(t) = 3t^2 \\ y'(t) = 2t \end{cases}$$

$$S = \int_1^3 \sqrt{(3t^2)^2 + (2t)^2} dt = S = \int_1^3 \sqrt{9t^4 + 4t^2} dt$$

$$= \sqrt{t^2(9t^2 + 4)} = \int_1^3 t \cdot \sqrt{9t^2 + 4} dt \rightarrow u = t^2$$

$$\int_1^3 t \cdot \sqrt{9u + 4} \frac{du}{2t} = \frac{1}{2} \int_1^3 \sqrt{9u + 4} du \quad \frac{du}{dt} = 2t \rightarrow dt = \frac{du}{2t}$$

$$v = 9u + 4 \rightarrow \frac{1}{18} \int_1^3 \sqrt{v} dv \rightarrow \frac{1}{18} \cdot \frac{2v^{\frac{3}{2}}}{\frac{3}{2}} = \frac{2v^{\frac{3}{2}}}{54} = \frac{v^{\frac{3}{2}}}{27}$$

$$\frac{dv}{du} = 9 \rightarrow du = \frac{dv}{9}$$

$$\frac{(9u + 4)^{\frac{3}{2}}}{27} = \frac{(9t^2 + 4)^{\frac{3}{2}}}{27} \rightarrow 3 \rightarrow \frac{(85)^{\frac{3}{2}}}{27} - \frac{(13)^{\frac{3}{2}}}{27} \text{ mC}$$

$$\rightarrow 1 \rightarrow \frac{(13)^{\frac{3}{2}}}{27}$$

$$\boxed{\text{ALTERNATIVA E}}$$

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⑤ Calcular a área delimitada pelo elipse:

$$\begin{cases} x = 3 \cos(t) \rightarrow -3 \sin(t) \\ y = \sin(t) \end{cases} \quad \text{INTERVALO} = \left[\frac{\pi}{2}, 0\right]$$

$$S = \int_{-3}^3 \sin(t) \cdot (-3 \sin(t)) dt = \int_{-3}^3 -3 \sin^2(t) dt = -3 \int_{-3}^3 \sin^2(t) dt$$

$$\rightarrow -3 \left[\frac{1}{2} (t - \sin(t) \cos(t)) \right] = -\frac{3t}{2} + \frac{3 \sin(t) \cos(t)}{2}$$

Resposta: 3π ua

ALTERNATIVA I

⑥ Calcular volume do sólido de revolução: $y = 2x^2$, $x = 1$, $x = 2$, $y = 2$, ao redor do eixo $y = 2$.

$$V = \pi \int_1^2 [2x^2 - 2] dx = \pi \int_1^2 4x^4 - 8x^2 + 4 dx \rightarrow 4\pi \int_1^2 x^4 - 2x^2 + 1$$

$$\int x^4 = \frac{x^5}{5}, \int -2x^2 = -\int 2x^2 = -\frac{2x^3}{3}, \int 1 = x$$

$$4\pi \left[\frac{x^5}{5} - \frac{8\pi x^3}{3} + 4\pi x + C \right] \Big|_1^2 =$$

$$2 \rightarrow \frac{4\pi \cdot 32}{5} - \frac{64\pi}{3} + 8\pi \rightarrow \frac{12\pi \cdot 32}{15} - \frac{320\pi}{15} + \frac{120\pi}{15} \rightarrow \frac{184\pi}{15}$$

$$1 \rightarrow \frac{4\pi}{5} - \frac{8\pi}{3} + 4\pi \rightarrow \frac{12\pi}{15} - \frac{40\pi}{15} + \frac{60\pi}{15} \rightarrow \frac{32\pi}{15}$$

$$\frac{184\pi}{15} - \frac{32\pi}{15} = \frac{152\pi}{15} \text{ uv.}$$

ALTERNATIVA II

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7) Calcule a área da superfície detida pela revolução do arco da parábola $y^2 = 8x$, $1 \leq x \leq 12$, ao redor do eixo dos x .

$$A = 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} dx$$

$$f'(x) = \sqrt{8} \cdot \sqrt{x} = \sqrt{8} \cdot x^{\frac{1}{2}} = \sqrt{8} \cdot \frac{x^{-\frac{1}{2}}}{2} = \frac{\sqrt{8}}{2\sqrt{x}}$$

$$[f'(x)]^2 = \frac{\sqrt{8}}{2\sqrt{x}} \cdot \frac{\sqrt{8}}{2\sqrt{x}} = \frac{8}{4x} = \frac{2}{x}$$

$$A = 2\pi \int_1^{12} \sqrt{8x} \sqrt{1 + \frac{2}{x}} dx = \int_1^{12} \sqrt{8x \cdot (1 + \frac{2}{x})} = \sqrt{8x + 16}$$

$$A = 2\pi \int_1^{12} \sqrt{8x + 16} dx = 2\pi \int_1^{12} \sqrt{u} \frac{du}{8} \Rightarrow \frac{\pi}{4} \int_1^{12} \sqrt{u} du \rightarrow$$

$$\rightarrow \frac{\pi}{4} \int_1^{12} u^{\frac{1}{2}} du = \frac{\pi}{4} \cdot \frac{2}{\frac{3}{2}} u^{\frac{3}{2}} \Big|_1^{12} = \frac{2\pi}{12} u^{\frac{3}{2}} \Big|_1^{12} = \frac{\pi}{6} u^{\frac{3}{2}} \Big|_1^{12}$$

$$u = 8x + 16$$

$$du = 8$$

$$dx \rightarrow dx = \frac{du}{8} \rightarrow \frac{\pi}{6} (8x + 16)^{\frac{3}{2}} + C$$

$$\rightarrow 12 \rightarrow \frac{\pi (112)^{\frac{3}{2}}}{6} - \frac{\pi (24)^{\frac{3}{2}}}{6}$$

$$\rightarrow 1 \rightarrow \frac{\pi (24)^{\frac{3}{2}}}{6}$$

ALTERNATIVA B

07/10/2012 $\rightarrow 120^\circ$

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8) a) $(-2, \frac{2\pi}{3})$

$$\frac{x}{2} = \cos(60^\circ) \Rightarrow x = -1, \quad \frac{y}{2} = \sin(60^\circ) \Rightarrow y = \frac{\sqrt{3}}{2} \cdot 2 = \sqrt{3}$$

PONTO: $(-1, \sqrt{3})$

ALTERNATIVA IV.

b) $(3, -\frac{\pi}{3}) \rightarrow 360^\circ - 60^\circ = 300^\circ$

$$\frac{x}{3} = \cos(60^\circ) \Rightarrow x = \frac{3}{2}$$

$$\frac{y}{3} = \sin(60^\circ) \Rightarrow y = -\frac{3\sqrt{3}}{2}$$

PONTO: $(\frac{3}{2}, -\frac{3\sqrt{3}}{2})$

ALTERNATIVA II.

9) Encontre (r, θ) , supondo $r < 0$ e $0 \leq \theta \leq 2\pi$ para o ponto P, cujas coordenadas são:

a) $(1, 1)$

$r = 1 \cdot \sqrt{2} = \sqrt{2}$ (Diagonal de quadrado).

$\theta = 45^\circ$ (bissetriz).

ou $\frac{\pi}{4}$

PONTO: $(\sqrt{2}, \frac{\pi}{4})$

ALTERNATIVA IV.

b) $(-1, 1)$

$\theta = 180^\circ - 45^\circ = 135^\circ = \pi - \frac{\pi}{4} = \frac{4\pi}{4} - \frac{\pi}{4} = \frac{3\pi}{4}$

$r = 1 \cdot \sqrt{2} = \sqrt{2}$

PONTO: $(-\sqrt{2}, \frac{3\pi}{4})$

ALTERNATIVA IV.

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1b) Achar o comprimento do arco da curva dado por $r = 1 + \cos(\theta)$

$$S = \int_{-\theta}^{\theta} \sqrt{f'(\theta)^2 + f(\theta)^2} d\theta \Rightarrow f'(\theta) = -\sin(\theta)$$

$$S = \int_0^{\pi} \sqrt{(-\sin(\theta))^2 + (1 + \cos(\theta))^2} d\theta \rightarrow$$

$$S = \int_0^{\pi} \sqrt{\sin^2(\theta) + 1 + 2\cos(\theta) + \cos^2(\theta)} d\theta \rightarrow$$

$$= \int_0^{\pi} \sqrt{2(1 + \cos(\theta))} d\theta = \int_0^{\pi} \sqrt{2} \sqrt{2\cos^2\left(\frac{\theta}{2}\right)} d\theta \rightarrow$$

$$\int_0^{\pi} 2\cos\left(\frac{\theta}{2}\right) d\theta = 2 \cdot 2\sin\theta \Big|_0^{\pi} = \frac{4\text{ u.c.}}{2} \quad \boxed{\text{ALTERNATIVA A.}}$$

1a) Calcular a área limitada pela curva $r = 5 + 2\sin(\theta)$.

$$A = \frac{1}{2} \int_0^{2\pi} r^2 d\theta = \frac{1}{2} \int_0^{2\pi} (5 + 2\sin(\theta))^2 d\theta \rightarrow$$

$$\frac{1}{2} \int_0^{2\pi} (25 + 20\sin(\theta) + 4\sin^2(\theta)) d\theta = \frac{1}{2} [25\theta - 20\cos(\theta) + 4$$

$$\int_0^{2\pi} \frac{1 - \cos(2\theta)}{2} d\theta] = -\frac{\theta}{2} - \frac{1}{4} \sin(2\theta)$$

$$\boxed{A = 27\pi \text{ u.c.}} \quad \boxed{\text{ALTERNATIVA B.}}$$